Individualized Cardiovascular Modeling for Operationally Relevant Altered-Gravity Scenarios

Hrudayavani S. Vellore [1, 2]
Texas A&M University, College Station, Texas, 77840, U.S.A.

Madison Herrmann [1]
Texas A&M University, College Station, Texas, 77840, U.S.A.

Cort Reinarz [1]
Texas A&M University, College Station, Texas, 77840, U.S.A.

Amanda E. Kraft [3]
Lockheed Martin, Cherry Hill, New Jersey, 08002, U.S.A.

Raquel C. Galvan-Garza [3]
Lockheed Martin, Cherry Hill, New Jersey, 08002, U.S.A.

Nancy Currie-Gregg [1]
Texas A&M University, College Station, Texas, 77840, U.S.A.

and

Ana Diaz-Artiles [1]
Texas A&M University, College Station, Texas, 77840, U.S.A.

Understanding individualized cardiovascular responses to acute altered gravity is critical for enhancing safety and performance of pilots and astronauts. Physiological responses vary substantially across individuals due to differences in sex, anthropometry, fitness level, and hydration status., yet experimentally isolating and quantifying these effects is costly and logistically challenging. This study uses a lumped-parameter computational cardiovascular model to quantify inter-individual variability in hemodynamic responses during two operationally relevant scenarios: G warm-up maneuver and parabolic flight. Model parameters were individualized using sex-specific anthropometric scaling relationships, including height, body mass, total blood volume, and body surface area, along with variations in athletic conditioning and hydration status. The framework evaluates heart rate (HR), stroke volume (SV), cardiac output (CO), cardiac index (CI), and arterial pressure responses under variable gravitational loading. Results show that while the qualitative cardiovascular response to gravitational transitions is consistent, the magnitude of these responses varies substantially between individuals. Larger individuals exhibit lower HR and higher SV, CO, and arterial pressure across gravitational phases. Dehydration increases HR and reduces CO and arterial pressure, reflecting reduced central blood volume. Athletic conditioning lowers HR, particularly in males, while stroke volume responses differ between sexes, with athletic males exhibiting higher SV and athletic females showing slightly lower SV compared with non-athletic counterparts. By enabling controlled comparison of individualized physiological characteristics under identical gravitational profiles, this computational framework provides a quantitative tool for characterizing inter-individual cardiovascular variability and supports the development of personalized monitoring strategies and countermeasure design for high-stress gravitational environments.

[1] Department of Aerospace Engineering, Texas A&M University, College Station, Texas, 77840, U.S.A
[2] Department of Aerospace and Mechanical Engineering, University of California, Davis, 95616, U.S.A.
[3] Lockheed Martin Advanced Technology Laboratories, Cherry Hill, New Jersey, 08002, U.S.A.

Acronyms and Nomenclature

BSA	=	Body Surface Area
C	=	Capacitance / Compliance
CI	=	Cardiac Index
CO	=	Cardiac Output
DBP	=	Diastolic Blood Pressure
F	=	Female
g	=	Gravity level
HR	=	Heart Rate
h	=	Height
LBNP	=	Lower Body Negative Pressure
LBPP	=	Lower Body Positive Pressure
LV	=	Left Ventricle
M	=	Male
m	=	Total body mass
MAP	=	Mean Arterial Pressure
P	=	Pressure
PCTL	=	Percentile
q	=	Flow
R	=	Resistance
r	=	Radius
SBP	=	Systolic Blood Pressure
SV	=	Stroke Volume
TBV	=	Total Blood Volume
V	=	Volume
ZPFV	=	Zero Pressure Filling Volume
$\rho_{\text{fat-free tissue}}$	=	Density of fat-free tissue

I. Introduction

As the pilot and astronaut populations diversify, it is essential to understand individual responses to altered-gravity environments. Humans experience numerous physiological changes outside of 1g, many of which are strongly influenced by hemodynamic fluid shifts resulting from both headward (negative inertial) and footward (positive) loads [1]. Spaceflight studies [2], [3] and ground-based analog testing [4], [5], [6] have demonstrated the magnitude of these fluid shifts under sustained reduced-gravity conditions. Such shifts can have critical implications for cardiovascular health, affecting both the severity of the response and the recovery time required to return to baseline after a maneuver. This research effort aims to characterize these health effects across a diverse population, accounting for differences height, weight, sex, cardiovascular profiles, and anthropometric characteristics.

A. Individual Cardiovascular Responses in Altered Gravity

Cardiovascular research in microgravity has examined both short- and long-term effects. Post-spaceflight stand tests suggest significant sex differences in autonomic responses, with higher incidence of syncope in females [7]. Females exhibit lower sympathetic nervous system activity in microgravity compared with males, evidenced by unchanged norepinephrine excretion [8], greater plasma volume loss [9], [10], reduced baroreflex control [9], [11], and lower orthostatic tolerance [12] following prolonged microgravity.

Under 1g conditions, females typically respond to physiological stress with increased heart rate (HR); whereas males rely more on increased vascular resistance [13], [14], [15], [16]. Convertino[17] reported that females exhibit less effective blood pressure regulation during orthostatic stress than males. Harm et al. [7] suggested that sex-dependent differences may be attenuated over long-term spaceflight, as observed in postflight tilt tests of the all-male Mir crew, which showed an 85% failure rate. These findings underscore the importance of continued investigation into sex differences under altered gravity.

Given the cost and logistical challenges of spaceflight studies, ground-based analogs are frequently used to simulate microgravity-induced cardiovascular changes. Head-down tilt (HDT) (6°) and head-down bed rest (HDBR) effectively replicate the cephalad fluid shift. Cowings et al. [18] reported substantial inter-individual differences in cardiovascular response during HDT, while Hordinsky et al. [19] found lower female orthostatic tolerance compared

with males during HDBR. Arzeno et al. [13] observed that females maintained blood pressure despite greater baroreflex sensitivity reduction, unlike males whose systolic/diastolic pressure decreased over time. Lower body positive pressure (LBPP) devices simulating microgravity fluid shifts indicated greater thoracic impedance and parasympathetic HR control in females, while males exhibited smaller increases in diastolic blood pressure [20].

While LBPP simulates microgravity, lower body negative pressure (LBNP) replicates orthostatic stress along the Gz axis. Whittle et al. [21] suggested that sex differences under LBNP are primarily driven by anthropometric variations, with marked differences in "absolute" variables (e.g., CO) but minimal differences in "indexed" variables (e.g., cardiac index, CI).

Collectively, these findings highlight the complex interplay between sex, autonomic regulation, and fluid distribution in both real and simulated microgravity. While ground-based analogs provide valuable insight into cardiovascular adaptation and orthostatic intolerance, inconsistencies across study designs and outcome measures underscore the need for standardized methodologies and sex-specific analyses. A deeper understanding of these physiological differences is essential for developing targeted countermeasures to mitigate cardiovascular deconditioning and optimize safety and performance across diverse individuals exposed to altered gravity.

B. Individualized Cardiovascular Modeling

Experimental studies of the effects of sex, athleticism, and body mass on cardiovascular responses in altered gravity environments are prohibitively expensive and time-consuming. Computational modeling provides an efficient and versatile alternative [22], [23], [24]. Such models have been widely used in medicine to simulate patient-specific responses [25]. Precision medicine, or patient-specific modeling, combines personal data, including sex, age, anthropometric parameters, and medical conditions, with physical equations governing cardiovascular function to create personalized simulations [26]. These models allow detailed examination of physiological responses to varying gravitational forces without requiring large experimental cohorts

The objective of this research effort is to develop a computational framework to evaluate the influence of key individual factors on cardiovascular responses during operationally relevant altered-gravity scenarios. Using established lumped-parameter cardiovascular modeling approaches, virtual subjects characterized by sex, height, body mass, hydration status, and athletic conditioning are simulated using a 21-compartment model. The framework quantifies cardiovascular variables across a range of gravitational conditions. These simulations provide insights into inter-individual differences in cardiovascular responses to acute altered-gravity. They also support the development of predictive algorithms and personalized countermeasures to prevent physiological impairments and optimize performance in diverse human populations, including operational groups such as pilots and astronauts.

II. Methods

A. Description of the Cardiovascular Model

The cardiovascular model used in this study is based on the model developed by Heldt [27] and subsequently expanded by Zamanian [28], Diaz-Artiles [29], and Whittle [30]. The 21-compartment lumped-parameter model employs a hydraulic-electric analogy to simulate the dynamics of the cardiovascular system. This type of model simplifies the complex behavior of the cardiovascular system by dividing it into discrete compartments, each representing key anatomical regions such as the heart chambers, arteries, veins, and capillaries. The model operates analogously to an electrical circuit, as pressures are analogous to voltages, flow to electrical current, resistances to vascular resistances, and compliances to capacitances. Model governing equations can be found below and in the appendix.

The model contains four body sections (head and arms, thorax, abdomen, and legs), encompassing 15 systemic vascular compartments (arterial and venous), 4 cardiac compartments, and 2 pulmonary compartments connected by a microvascular resistance (assumed linear). The architecture is shown in Fig. 1.

Each compartment *n* is characterized by a resistance R_n and compliance C_n, relating distending volume $V_{d,n}$ to the transmural pressure $P_n - P_{ext,n}$, where $P_{ext,n}$ represents external pressures (e.g., intrathoracic, intra-abdominal, or muscle pump) and $P_{h,n}$ represents gravitational hydrostatic pressure. The flow between compartments is given by:

$$q_n = \frac{P_{n-1} - P_n - P_{h,n}}{R_n} \tag{1}$$

For most compartments, the transmural pressure is assumed to vary linearly with distending volume:

$$P_n - P_{ext,n} = \frac{V_{d,n}}{C_n} \tag{2}$$

Compartments 11 (splanchnic), 13 (leg), and 14 (abdominal) instead follow a nonlinear pressure-volume relationship to capture their behavior at high transmural pressures:

$$V_{t,n} = V_{0,n} + \frac{2V_{max,n}}{\pi} \cdot \arctan\left(\frac{C_{0,n}}{2V_{max,n}} \cdot P_n\right), P > 0 \quad (3)$$

where $V_{max,n}$ defines the distending volume limit and $C_{0,n}$ is the compliance at zero transmural pressure. The full model, implemented in MATLAB Simulink, consists of coupled first-order differential equations with distending volume as the state variable:

$$\frac{d}{dt}V_{d,n}(t) = q_{in,n} - q_{out,n} \quad (4)$$

The four cardiac chambers (left and right atria and ventricles) are modeled using time-varying elastance, cycling between low-capacitance (systolic) and high-capacitance (diastolic) states. Heartbeat generation is represented via an integral pulse frequency modulation model, with intrathoracic pressure modulating cardiac filling. Cardiac valves are represented as unidirectional diodes to prevent flow reversal, and venous valves in compartments 4 (upper body) and 13 (leg) are similarly represented.

Short-term cardiovascular regulation under orthostatic or other stress is governed by two neurally mediated feedback systems. The arterial baroreflex uses a lumped carotid sinus baroreceptor located 25 cm cranial of the heart, with carotid sinus pressure computed as aortic arch pressure minus the orthostatic-induced hydrostatic column. Deviations from the arterial pressure set-point are passed through sympathetic and parasympathetic linear time-invariant filters, with effector-specific gains applied to HR, ventricular contractility, peripheral resistance, and venous unstressed volume across the upper body, renal, splanchnic, and leg compartments. The cardiopulmonary reflex operates analogously, using right atrial transmural pressure relative to a set-point, with sympathetic filter gains modulating the same peripheral resistance and venous unstressed volume effectors. The total neurally mediated contribution to each effector variable is the sum of both reflex systems.

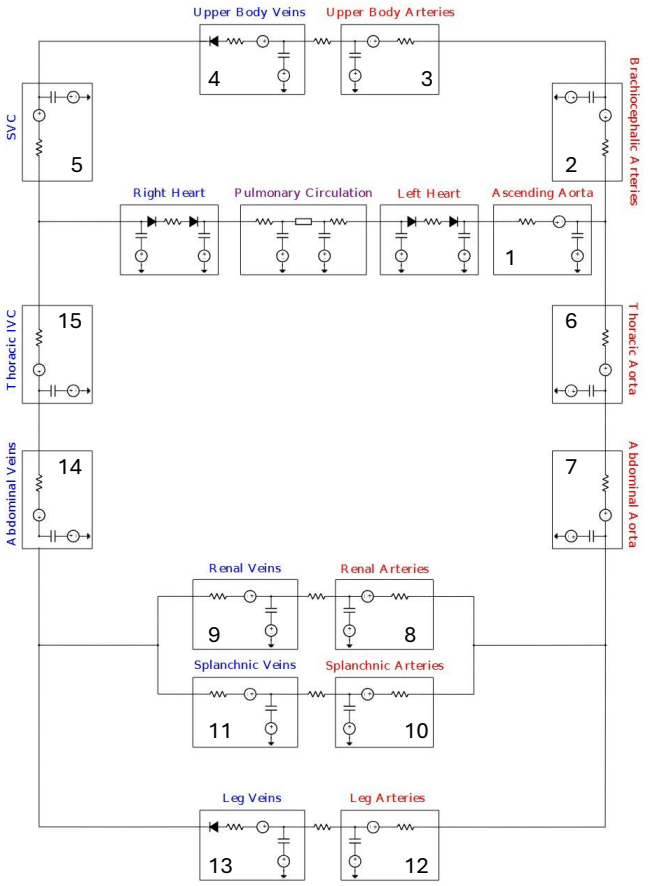


Figure 1. Simplified diagram of the cardiovascular model.

B. Definition of Simulated Individuals

Several parameters were selected to individualize the cardiovascular model, including height, total body mass, total blood volume (TBV), body surface area, athletic ability, and hydration status. Simulated individuals were defined by sex and include the following categories: 5th percentile (PCTL), 50th PCTL, 95th PCTL, athletic 50th PCTL, and dehydrated 50th PCTL [31]. Height and total mass data used to determine percentiles were obtained from the 2012 anthropometric study of U.S. Army personnel [31]. Specific values for all the parameters for each simulated individual are shown in Table 1, and the rationale for selecting these parameters is described in the following paragraphs.

An individual's height has been shown affect HR [32] in a 1g environment, while total body mass influences HR variability [33]. Prior work has demonstrated that cardiovascular responses can differ between anthropometrically distinct males and females [34]. TBV was estimated using the method developed by Feldschuh et al. [35], which accounts for sex differences in blood volume. This approach uses height and total body mass data from the anthropometric study, ensuring that, for instance, a male and female of the same height and mass have different TBV values. Additionally, the maximum fluid volume allowed to shift into the interstitial space (V_{max}, defined as 350 mL by Zamanian [28] and Diaz Artiles [29]), and the non-stressed vascular volume (zero pressure filling volume, ZPFV)

of each compartment were proportionally scaled to match each individual's TVB based on their anthropometry. The V_{max} for each individual is reported in Table 1, and the ZPFVs are presented in Table 2.

Body surface area (BSA) is used to calculate CI from CO, where CI = CO/BSA. BSA is estimated from total body mass (m) and height (h) using the sex-specific equations provided by Tikuisis et al. [36], as shown below:

$$\text{BSA} = 128.1 * m^{0.44} * h^{0.60} \quad \text{(males)} \qquad \text{BSA} = 147.4 * m^{0.47} * h^{0.55} \quad \text{(females)} \qquad (5)$$

Today, warfighters operate in increasingly extreme environments, making it critical to understand the effects of dehydration under the high gravitational forces experienced by high-performance pilots. In the cardiovascular system, dehydration reduced both TBV and plasma volume [37], leading to decreased stroke volume (SV) and compensatory increases in HR to maintain CO [38]. Following the approach of Dill and Costill [39], dehydration was simulated by reducing TBV by 9.6%. Variations in TBV are well-established to influence cardiovascular responses under altered gravity conditions [40].

Athletic ability significantly influences baseline cardiovascular parameters in a 1-g environment, with the magnitude of the changes depending on the type of conditioning [41]. Pelliccia et al. [42] studied left ventricular (LV) dimensions of 1,309 elite athletes across 38 sports, finding particularly dilated LV chambers (> 60 mm) in individuals with larger body mass and endurance training. LV end-diastolic dimensions ranged from 38 to 66 mm for females and 43 to 70 mm for males. Pelliccia e al. [42] also noted that resting HRs are generally lower in endurance athletes. The LV dimensions and baseline HRs used in this study are derived from these findings. Athletic ability is further characterized by LV elasticity, which is known to change with exercise [43], with relevant values taken from Starling et al [44].

Parameters	5th PCTL F	5th PCTL M	50th PCTL F	50th PCTL M	95th PCTL F	95th PCTL M	50th PCTL Dehydrated F	50th PCTL Dehydrated M	50th PCTL Athletic F	50th PCTL Athletic M
TBV [mL]	3642.3	4508.0	4275.2	5499.0	5051.8	6586.7	3864.8	5050.3	4275.2	5499.0
BSA [m^2]	1.490	1.730	1.747	2.036	2.054	2.392	1.747	2.036	1.747	2.036
Baseline HR [bpm]	73.2	70.3	73.2	70.3	73.2	70.3	73.2	70.3	69.5	65.4
Height [cm]	152.5	164.8	162.6	175.5	174.0	187.0	162.6	175.5	162.6	175.5
Total Mass [kg]	51.3	64.4	66.8	84.6	87.1	110.7	66.8	84.6	66.8	84.6
Height of Head [cm]	22.0	23.3	24.0	25.4	26.0	27.5	24.0	25.4	24.0	25.4
Upper Arm Length [cm]	28.3	30.7	31.1	33.5	34.0	36.5	31.1	33.5	31.1	33.5
Forearm Length [cm]	21.8	24.4	24.0	26.7	26.8	29.5	24.0	26.7	24.0	26.7
Seated Leg Length [cm]	73.2	78.0	79.4	84.7	86.0	91.7	79.4	84.7	79.4	84.7
Tibial Height [cm]	39.9	42.6	43.7	46.7	47.9	51.3	43.7	46.7	43.7	46.7
Left Ventricle ZPFV [mL]	32.04	39.65	49.8	48.37	44.44	57.94	37.61	48.37	54.8	59.0
LV Contractility [cm]	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.15	0.15
V_{max} [mL]	247.5	306.4	290.6	373.7	343.3	447.6	290.6	373.7	290.6	373.7

Table 1. Individualized input parameters. *Input parameters for the 5th percentile (PCTL) Female (F), 5th PCTL Male (M), 50th PCTL F, 50th PCTL M, 95th PCTL F, 95th PCTL M, 50th PCTL Dehydrated F, 50th PCTL Dehydrated M, 50th PCTL Athletic F, and 50th PCTL Athletic M. Highlighted values are the parameters that differ from the 50th PCTL female and 50th PCTL male. TBV: total blood volume; BSA: body surface area; HR: heart rate; ZPFV: zero pressure filling volume; LV: left ventricle; V_max: maximum volume of fluid permitted to shift to the interstitial space.*

Parameters [mL]	5th PCTL F	5th PCTL M	50th PCTL F	50th PCTL M	95th PCTL F	95th PCTL M	50th PCTL Dehydrated F	50th PCTL Dehydrated M	50th PCTL Athletic F	50th PCTL Athletic M
ZPFV1	14.85	18.38	17.43	23.80	20.60	26.86	17.43	23.80	17.43	23.80
ZPFV2	3.54	4.38	4.15	5.67	4.90	6.39	4.15	5.67	4.15	5.67
ZPFV3	141.45	175.07	166.03	226.70	196.19	255.79	166.03	226.70	166.03	226.70
ZPFV4	456.17	564.59	535.44	731.09	632.70	824.93	535.44	731.09	535.44	731.09
ZPFV5	11.32	14.01	13.28	18.14	15.69	20.46	13.28	18.14	13.28	18.14

Parameters [mL]	5th PCTL F	5th PCTL M	50th PCTL F	50th PCTL M	95th PCTL F	95th PCTL M	50th PCTL Dehydrated F	50th PCTL Dehydrated M	50th PCTL Athletic F	50th PCTL Athletic M
ZPFV6	11.32	14.01	13.28	18.14	15.69	20.46	13.28	18.14	13.28	18.14
ZPFV7	7.07	8.75	8.30	11.33	9.81	12.79	8.30	11.33	8.30	11.33
ZPFV8	14.14	17.51	16.60	22.67	19.62	25.58	16.60	22.67	16.60	22.67
ZPFV9	21.22	26.26	24.90	24.90	29.43	38.37	24.90	24.90	24.90	24.90
ZPFV10	212.17	262.6	249.04	340.04	294.28	383.69	249.04	340.04	249.04	340.04
ZPFV11	810.50	1003.14	951.34	1298.70	1124.15	1465.69	951.34	1298.70	951.34	1298.70
ZPFV12	141.45	175.07	166.03	226.70	196.19	255.79	166.03	226.70	166.03	226.70
ZPFV13	506.39	626.75	594.38	811.57	702.35	915.74	594.38	811.57	594.38	811.57
ZPFV14	55.87	69.15	65.58	89.54	77.49	101.04	65.58	89.54	65.58	89.54
ZPFV15	23.34	28.89	27.39	37.40	32.37	42.21	27.39	37.40	27.39	37.40

Table 2. Zero pressure filling volumes for all individuals. *Zero pressure filling volumes (ZPFV) of the 15 systemic compartments for the 5th percentile (PCTL) Female (F), 5th PCTL Male (M), 50th PCTL F, 50th PCTL M, 95th PCTL F, 95th PCTL M, 50th PCTL Dehydrated F, 50th PCTL Dehydrated M, 50th PCTL Athletic F, and 50th PCTL Athletic M. Highlighted values are the parameters that differ from the 50th PCTL female and 50th PCTL male. Values for each compartment were proportionally scaled to match each individual's total blood volume based on their anthropometry.*

1. Modeling the Seated Pilot

To accurately simulate the seated posture in the cockpit, the computational model was modified in three key ways (see Fig. 2): 1) Leg compartment lengths were shortened to reflect the seated posture; 2) Body tilt of 30° was applied to match the cockpit seat angle; and 3) External pressure was applied to the lower body compartments to simulate compression experienced in high-G environments. These adjustments are illustrated in Fig. 2and described in detail in the following paragraphs.

Leg compartment length: The length of the leg compartments was set equal to the height of the shin (h_{shin}) plus twice the radius of the thigh (r_{thigh}):

$$Leg\ Compartment\ Length = 2r_{thigh} + h_{shin} \quad (6)$$

This reduces the effective length of the lower limbs by excluding the thigh height, thereby decreasing the gravitational load on the legs.

Body tilt: Simulated individuals were tilted back by 30° to replicate the seat configuration in an F-16 cockpit [45]. In the seated posture, the vertebral column experiences the greatest stress during hypergravity. The 30° recline mitigates spinal injuries and neck strain by reducing the gravitational load along the Gz axis [45].

Lower-body compression: To simulate compression on the lower limbs while seated, external pressure was applied to the leg compartments. The magnitude of this pressure was calculated using the hydrostatic pressure equation:

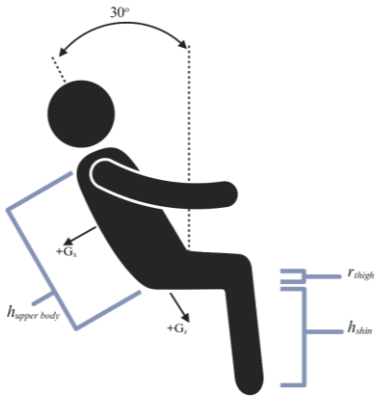


Figure 2. Adjustments to the computational model to represent the seated cockpit position. *Anthropometric parameters used to define this configuration are indicated: h_{shin}: shin height; r_{thigh}: thigh radius; $h_{upper\ body}$: upper body height. The body-fixed $+G_z$ and $+G_x$ axes are also shown.*

$$P_{ext}[mmHg] = \rho_{fat-free\ tissue} \times g \times \frac{2}{3}\left(h_{upper\ body} + r_{thigh}\right) \quad (7)$$

where $\rho_{fat-free\ tissue}$ is the density of fat-free tissue (1.10 g/cm^3), g is the gravity level, $h_{upper\ body}$ is the height of the upper body, and r_{thigh} is the radius of the thigh, as illustrated in Fig. 2.

C. Operationally Relevant Scenarios

Two operationally relevant flight scenarios were simulated for their relevance in pilot training and performance in altered-gravity conditions: G warm-up and parabolic flight (Fig. 3).

1. G Warm-Up

The G warm-up maneuver is a standard pilot training procedure [46]. In this maneuver, the pilot performs a 180° turn, generating 3 g along the body's +Gz-axis. The turn is repeated in both directions (left and right). The sequence begins with 1 g held for 1 s, followed by a 3 s transition to 3 g. The pilot maintains 3 g for 10 s, then transitions back to 1 g over 3 s. The 1g phase is held for 5 s before another 3 s ramp-up to 3 g, maintained again for 10 s, and finally returning to 1g over 3 s. The full gravity profile is depicted in Fig. 3 (left).

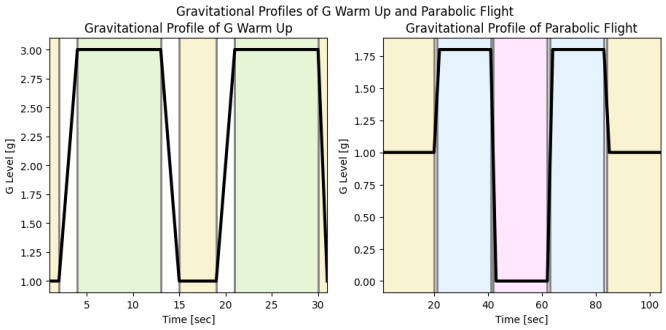


Figure 3. Illustrations of the two operationally relevant scenarios simulated. *G warm-up maneuver (left) and parabolic flight profile (right).*

2. Parabolic Flight

Parabolic flight maneuvers have been extensively used to study the effects of acute gravitational changes on human physiology [47], [48], as they provide gravitational profiles similar to those experienced by high-performance pilots. The simulated parabolic flight scenario consists of the following sequence: 20 s at 1 g, followed by a 1 s transition to 1.8 g, which is maintained for 20 s. The aircraft then transitions over 1 s to 0 g, which is held for 20 s. A subsequent 1 s transition returns gravity to 1.8 g, maintained for 20 s, followed by a final 1 s transition back to 1 g, which is held for the last 20 s. The full gravity profile is depicted in Fig. 3 (right).

III. Results

A. Cardiovascular Response to a G Warm-Up Maneuver

Fig. 4 illustrates the impact of anthropometric differences on cardiovascular responses during the G warm-up scenario. Across individuals, gravity-dependent trends were consistent, with differences in magnitude primarily driven by anthropometry. Importantly, the simulations allow quantitative resolution of these inter-individual differences under identical gravitational loading conditions.

HR increases sharply during the first hypergravity (3 g) phase and decreases during the subsequent normogravity (1g) phase as the system approaches a new steady state. During the second 3 g exposure, HR rises again.

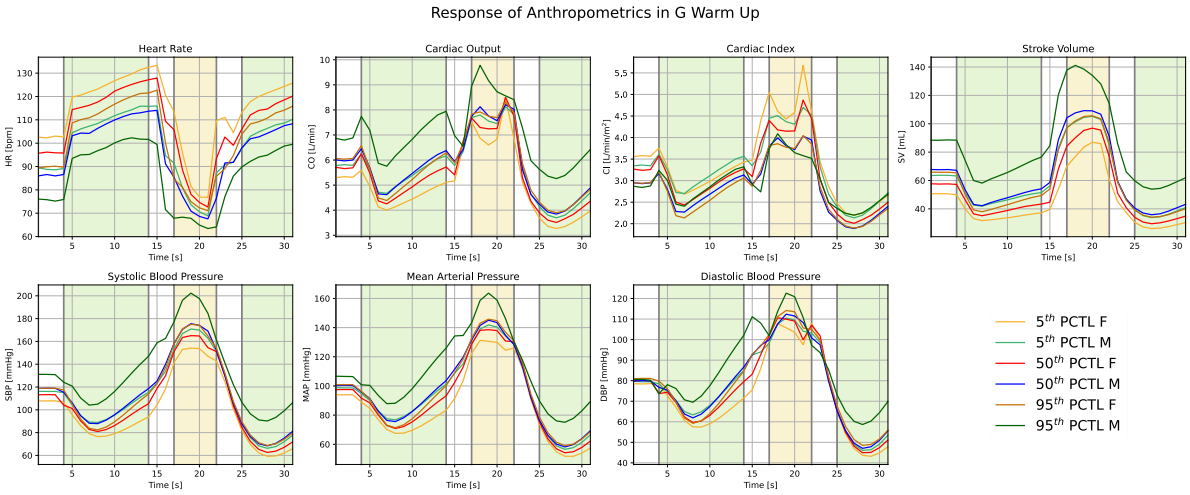


Figure 4. Cardiovascular responses during the G warm-up maneuver for different anthropometric groups. *Variables include heart rate (HR), cardiac output (CO), cardiac index (CI), stroke volume (SV), systolic blood pressure (SBP), mean arterial pressure (MAP), and diastolic blood pressure (DBP)***.** *The figure illustrates the impact of gravitational forces on individuals of varying body sizes, highlighting distinct cardiovascular responses across anthropometric percentiles. PCTL: percentile; M: male; F: female. Green background: hypergravity (3 g); yellow background: normogravity (1 g).*

Anthropometrically smaller individuals (e.g., 5th PCTL female) exhibit higher HR across all gravity phases, whereas larger individuals (e.g., 95th PCTL male) demonstrate consistently lower HR.

SV decreases during the first 3 g phase relative to the initial 1 g baseline, reflecting reduced central venous return under hypergravity. During the subsequent 1 g phase, SV increases, briefly exceeding steady-state 1 g levels before decreasing again during the second 3 g phase. Larger individuals, particularly the 95th PCTL male, maintain higher SV values across all phases compared with smaller individuals. This is consistent with their greater TBV and larger cardiac dimensions, which support higher absolute SV during gravitational stress. The inter-individual variability in SV underscores the key role of anthropometric factors on the ability to maintain CO.

CO follows trends similar to SV, decreasing during each 3 g phase and recovering during 1 g. Larger individuals consistently demonstrate higher absolute CO across all phases, corresponding to their higher SV. CI, however, demonstrates a different pattern. During the initial 1 g phase, CI is inversely related to body size, with smaller individuals exhibiting higher values. This reflects both normalization to a smaller body surface area and the higher baseline HR observed in smaller individuals in the simulation. As the maneuver progresses, this ordering becomes less distinct, suggesting that compensatory responses to gravitational stress are not strictly proportional to body size and may involve differential peripheral resistance adjustments and blood volume redistribution. Mean arterial pressure (MAP) decreases during the first 3 g phase and increases during the subsequent 1 g period, followed by another decline during the second hypergravity phase. The 95th PCTL male exhibits the highest absolute MAP across all phases, potentially reflecting greater vascular capacity and blood volume supporting more stable arterial pressure regulation. Similar qualitative trends are observed for systolic (SBP) and diastolic blood pressure (DBP). Collectively, these results demonstrate that while the direction of cardiovascular responses to hypergravity is predictable, their magnitude varies systematically with anthropometry. The modeling framework enables quantitative characterization of these differences, providing a basis for incorporating individual-specific parameters into predictive monitoring and countermeasure algorithms.

Fig. 5 illustrates the cardiovascular responses of hydrated and dehydrated 50th PCTL individuals during the G warm-up maneuver. Dehydration significantly alters cardiovascular regulation across all gravity phases. HR is consistently elevated in the dehydrated condition, reflecting compensatory increases in HR in response to reduced central blood volume. SV is markedly reduced, consistent with diminished preload secondary to decreased TBV. Although SV increases during the second 1 g phase in both conditions; the magnitude of recovery is attenuated in dehydrated individuals. Consequently, CO and CI are lower in the dehydrated state despite the elevated HR, indicating incomplete compensation. SBP, DBP, and MAP are also reduced throughout the gravity profile, suggesting impaired ability to maintain arterial pressure under combined hypovolemic and gravitational stress. While the qualitative effects of dehydration are well established, these simulations quantify the magnitude of performance degradation during

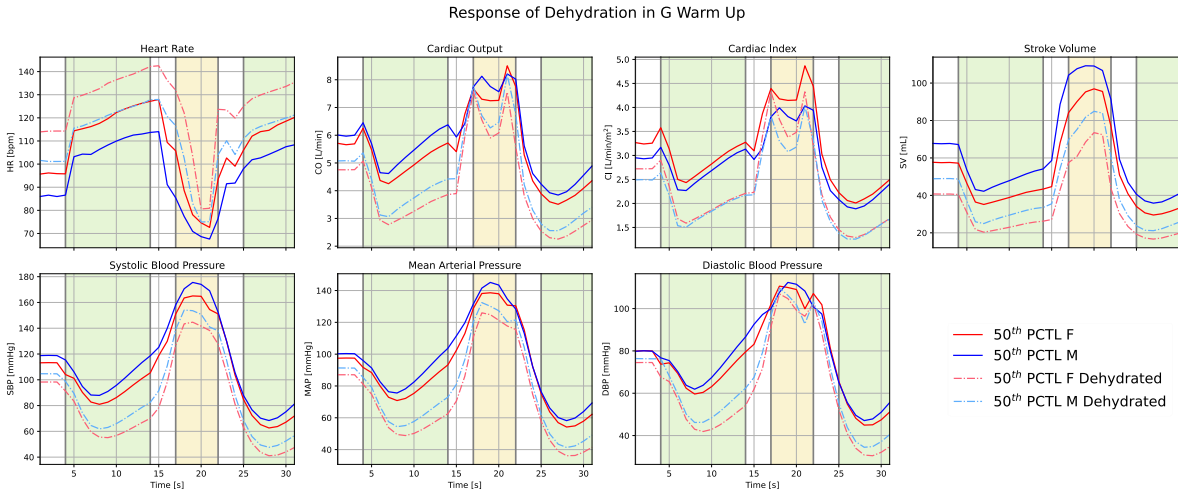


Figure 5. Cardiovascular responses during the G warm-up maneuver for hydrated and dehydrated individuals. *Shown are heart rate (HR), cardiac output (CO), cardiac index (CI), stroke volume (SV), systolic blood pressure (SBP), mean arterial pressure (MAP), and diastolic blood pressure (DBP) for 50th percentile (PCTL) males and females under both hydration conditions. PCTL: percentile; M: male; F: female. Green background: hypergravity (3 g); yellow background: normogravity (1 g).*

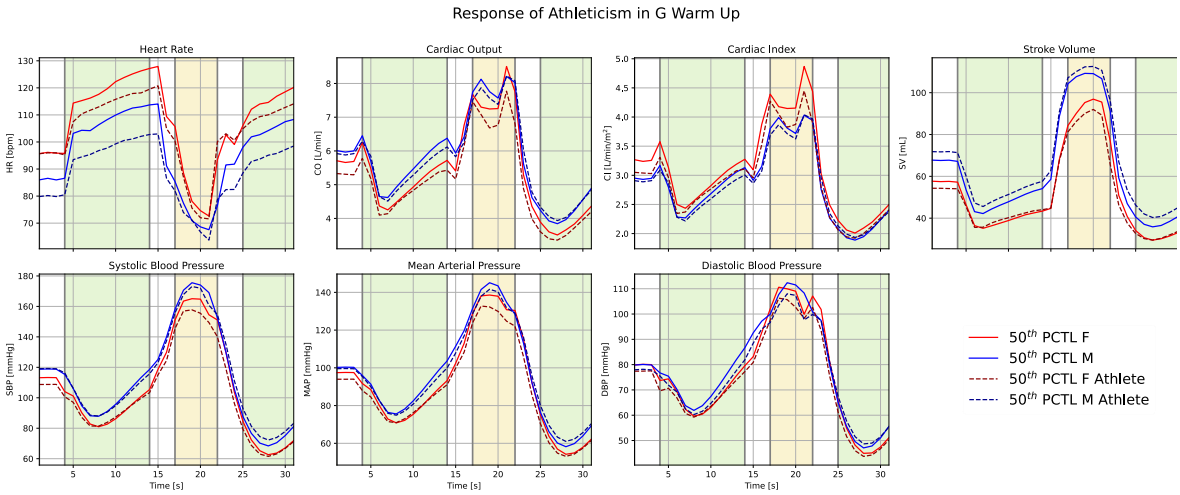


Figure 6. Cardiovascular responses during the G warm-up maneuver for athletic individuals. *Variables include heart rate (HR), cardiac output (CO), cardiac index (CI), stroke volume (SV), systolic blood pressure (SBP), mean arterial pressure (MAP), and diastolic blood pressure (DBP) for the 50th percentile (PCTL) male and female athletes compared with their non-athletic counterparts. PCTL: percentile; M: male; F: female. Green background: hypergravity (3 g); yellow background: normogravity (1 g).*

dynamic gravitational transitions. Such quantification is essential for developing thresholds, risk models, and adaptive decision-support systems that account for hydration status in operational environments.

Fig. 6. Illustrates the cardiovascular responses during the G warm-up maneuver for 50th percentile (PCTL) non-athletic and athletic individuals. Athletic conditioning is associated with a significantly lower HR across all gravity phases, reflecting improved cardiovascular efficiency. Despite the lower HR, CO, CI, and arterial pressures (systolic, diastolic, and mean) remain comparable between groups, indicating preserved hemodynamic stability during hypergravity exposure. These findings suggest that athletic individuals tolerate gravitational loading while

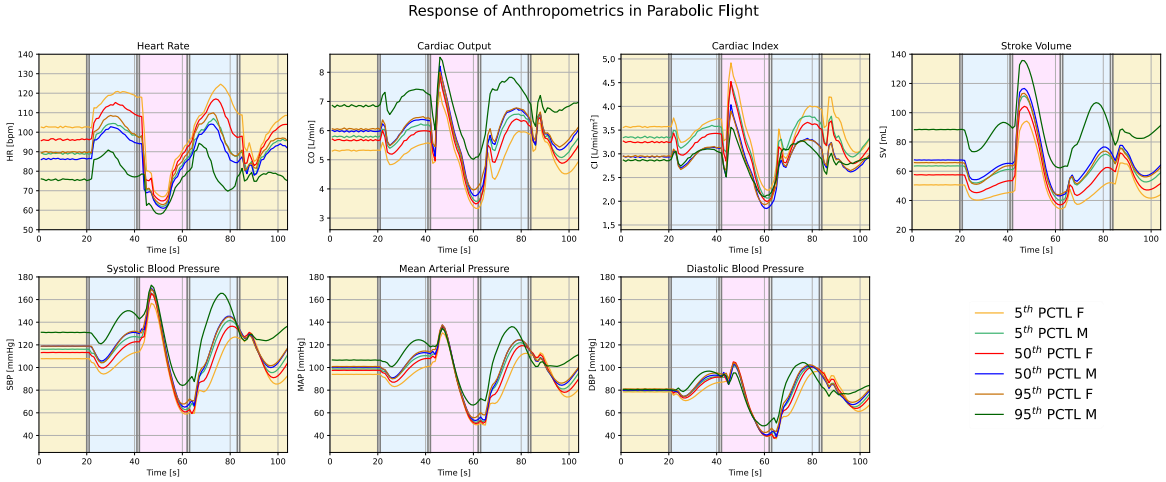


Figure 7. Cardiovascular responses during the parabolic flight maneuver for different anthropometric groups. *Variables include heart rate (HR), cardiac output (CO), cardiac index (CI), stroke volume (SV), systolic blood pressure (SBP), mean arterial pressure (MAP), and diastolic blood pressure (DBP). The figure illustrates the impact of gravitational forces on individuals of varying body sizes, highlighting distinct cardiovascular responses across different anthropometric percentiles. PCTL: percentile; M: male; F: female. Yellow background: normogravity (1 g); blue background: hypergravity (1.8 g); pink background: microgravity (0 g).*

maintaining adequate cardiovascular performance at a lower cardiac workload. Beyond confirming expected physiological trends, the model enables quantitative comparison of conditioning-related differences under identical stress profiles, supporting the potential integration of fitness-related parameters into individualized performance prediction and countermeasure design.

B. Cardiovascular Response to a Parabolic Flight Maneuver

Fig. 7 illustrates the cardiovascular responses of individuals with varying anthropometric profiles during the simulated parabolic flight maneuver. Across individuals, the qualitative pattern of response to the gravitational phases (1 g, 1.8 g, and 0 g) remain consistent, although the magnitude of the responses varies with body size. Smaller individuals, such as the 5th PCTL female, exhibit higher HR across all phases compared to larger individuals, with the 95th PCTL male displaying the lowest HR.

SV also varies systematically with anthropometry. Larger individuals, particularly the 95th PCTL male, maintain higher SV across all gravitational phases compared with smaller individuals such as the 5th PCTL female. These differences persist during both hypergravity and microgravity phases, with smaller individuals exhibiting consistently lower SV. CO follows trends similar to SV, with larger individuals maintaining higher absolute CO throughout the maneuver. When normalized to body surface area, however, cardiac index (CI) demonstrates the opposite pattern: smaller individuals maintain higher CI values across the maneuver, reflecting both their smaller body surface area and higher HR. MAP also shows systematic differences across anthropometric groups. Larger individuals maintain higher MAP values across all gravity phases, with these differences being most evident during hypergravity. Similar trends are observed for SBP and DBP. Interestingly, the differences in HR, CO, and MAP between individuals are least pronounced during the microgravity phase, suggesting that the reduction in hydrostatic gradients partially attenuates anthropometry-dependent cardiovascular differences.

Together, these results demonstrate that although the direction of cardiovascular responses to rapid gravitational transitions is consistent across individuals, the magnitude of these responses varies substantially with anthropometric characteristics. The simulation framework enables quantitative characterization of these inter-individual differences, providing a basis for incorporating anthropometric parameters into predictive models of cardiovascular performance during dynamic gravity profiles.

Fig. 8 displays cardiovascular responses during parabolic flight while highlighting the effects of hydration status. Similar to the findings observed during the G warm-up maneuver, dehydration results in elevated HR across all gravity phases compared with the hydrated condition. SV is markedly reduced in dehydrated individuals, reflecting reduced preload associated with decreased total blood volume. As a consequence, both CO and CI are reduced under dehydrated conditions, with the most pronounced reduction occurring during hypergravity phases. Dehydration also

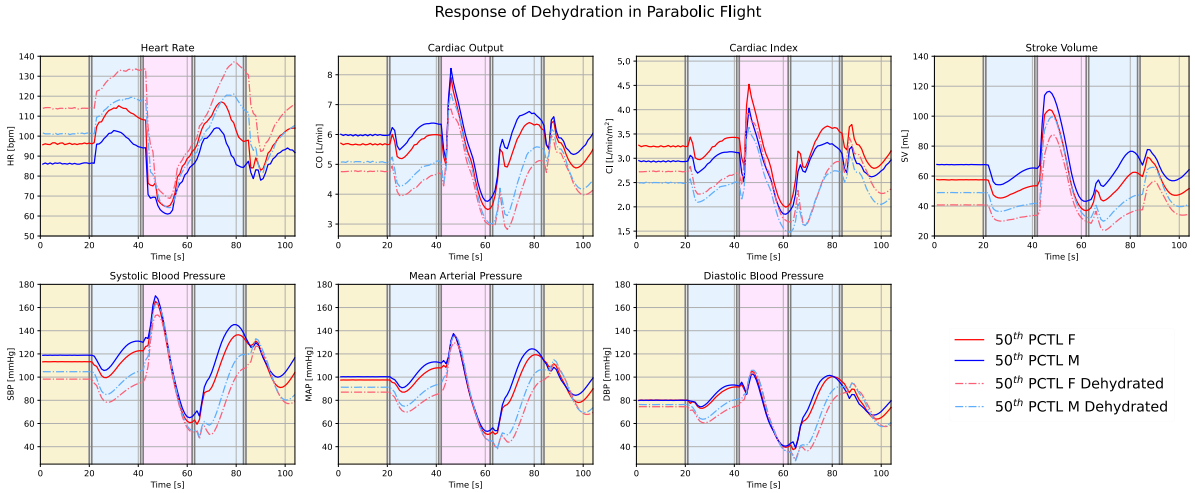


Figure 8. Cardiovascular responses during the parabolic flight maneuver for hydrated and dehydrated individuals. *Shown are heart rate (HR), cardiac output (CO), cardiac index (CI), stroke volume (SV), systolic blood pressure (SBP), mean arterial pressure (MAP), and diastolic blood pressure (DBP) for 50th percentile (PCTL) males and females under both hydration conditions. PCTL: percentile; M: male; F: female. Yellow background: normogravity (1 g); blue background: hypergravity (1.8 g); pink background: microgravity (0 g).*

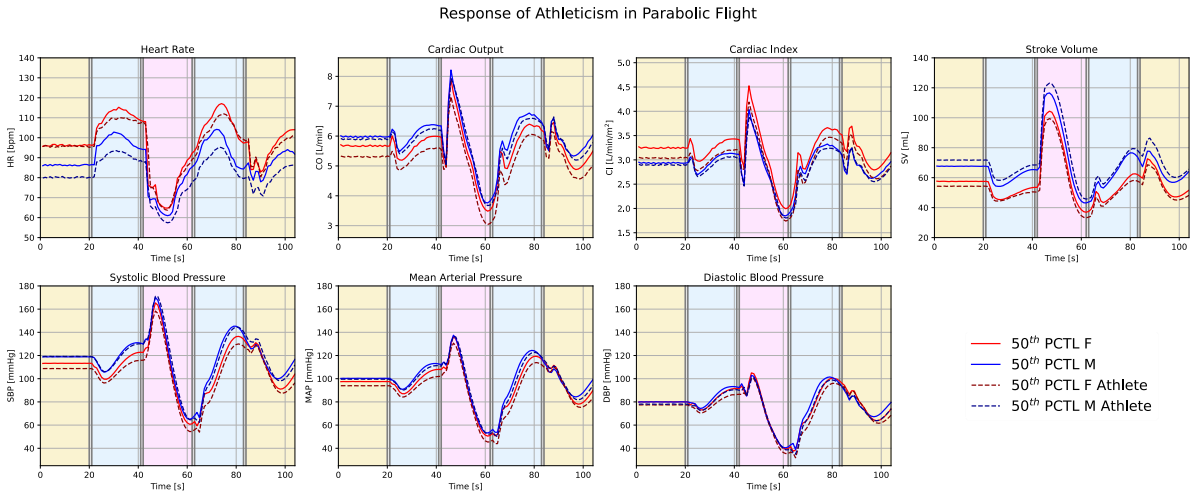


Figure 9. Cardiovascular responses during the parabolic flight maneuver for athletic individuals. *Variables include heart rate (HR), cardiac output (CO), cardiac index (CI), stroke volume (SV), systolic blood pressure (SBP), mean arterial pressure (MAP), and diastolic blood pressure (DBP) for the 50th percentile (PCTL) male and female athletes compared with their non-athletic counterparts. PCTL: percentile; M: male; F: female. Yellow background: normogravity (1 g); blue background: hypergravity (1.8 g); pink background: microgravity (0 g).*

results in lower SBP, DBP, and MAP across the entire maneuver, suggesting a reduced capacity to maintain arterial pressure under combined hypovolemic and gravitational stress.

While the qualitative physiological effects of dehydration are well understood, these simulations provide quantitative estimates of how dehydration alters cardiovascular responses during rapid gravity transitions. Such quantification is essential for developing individualized monitoring strategies and operational thresholds that account for hydration status during dynamic flight profiles.

Fig. 9 illustrates the cardiovascular responses during parabolic flight for 50th percentile (PCTL) non-athletic and athletic individuals. Across all gravity phases, athletic individuals demonstrate a lower HR compared with their non-athletic counterparts, especially males, consistent with improved cardiovascular efficiency associated with athletic conditioning. Differences in SV between athletic and non-athletic individuals vary slightly between sexes in the simulation, with the athletic male demonstrating a modestly higher SV compared with his non-athletic counterpart, while the athletic female exhibits a slightly lower SV. These differences translate into small variations in CO. In particular, the athletic female shows lower CO compared with the non-athletic female during the initial 1 g phase and during both hypergravity phases. Despite these differences, arterial pressures (SBP, DBP, and MAP) remain broadly comparable between athletic and non-athletic individuals across the maneuver, indicating preserved overall hemodynamic regulation during parabolic flight. These findings underscore the complex interactions between athletic conditioning and cardiovascular responses during rapid gravitational transitions and highlight potential sex-based differences in how conditioning influences cardiovascular performance. More broadly, the simulation framework enables quantitative comparison of these interactions under identical gravitational stress profiles, supporting further investigation into their implications for performance in high-stress environments and for the development of individualized predictive models and countermeasure strategies.

IV. Conclusion

This study advances understanding of individualized cardiovascular responses to altered gravitational forces by quantifying the influence of anthropometric characteristics, athletic conditioning, and hydration status on physiological stability during dynamic gravity profiles. Across both simulated scenarios, individuals exhibited consistent qualitative cardiovascular responses to gravitational transitions; however, the magnitude of these responses varied substantially across individuals. Smaller individuals consistently exhibited higher HR and lower SV during hypergravity phases, whereas larger individuals maintained higher SV and more stable arterial pressure. These findings highlight the importance of considering anthropometric variability when evaluating cardiovascular tolerance to gravitational stress.

Body size and cardiovascular dynamics are therefore critical parameters for optimizing performance in variable gravitational environments. Our findings support observations reported in the literature indicating that individuals with lower TBV compensate for reduced SV through increased HR to maintain CO, while larger individuals maintain more stable MAP through greater SV [49]. By quantifying these differences under identical gravitational profiles, the modeling framework provides a tool for evaluating individualized countermeasure strategies, including anthropometrically informed compression garments or anti-G suit configurations tailored to specific pilot characteristics.

Athletic conditioning was associated with lower HR across all gravitational phases, suggesting improved cardiovascular efficiency during gravitational stress. Considering fitness level when designing training protocols and operational countermeasure may therefore improve tolerance to repeated gravitational loading in both aviation and spaceflight environments. Physically fit individuals often demonstrate enhanced autonomic regulation and baroreflex sensitivity, which may help mitigate cardiovascular strain during rapid gravitational transitions [50]. Similarly, dehydration produced measurable reductions in HR, SV, and arterial pressure across scenarios, reinforcing the importance of hydration management strategies, particularly in operational settings where fluid redistribution and dehydration can occur.

A primary limitation of this work is the absence of direct experimental validation of the simulated G warm-up and parabolic flight scenarios. However, the underlying lumped-parameter cardiovascular model has been previously validated across multiple gravitational and orthostatic stress conditions [24], [27], [28], [30], [40]. Future work should incorporate experimental datasets from centrifugation or parabolic flight campaigns to further refine and validate individualized model predictions.

Overall, this work demonstrates that substantial inter-individual variability can arise in cardiovascular responses to gravitational stressors. By enabling quantitative characterization of these differences, individualized cardiovascular modeling provides a foundation for incorporating personal physiological parameters into predictive monitoring systems, adaptive countermeasure design, and human–machine integration strategies. Such approaches may ultimately improve operational performance, enhance safety margins, and support more resilient human performance in high-stress gravitational environments.

Appendix

The equations for the model are adapted from a model built initially by Heldt [27] and compounded on by Zamanian [28] and Diaz Artiles [29]. They are included here for completeness.

A. Cardiac Model

The elastance of the cardiac chambers is time varying. In the equation below, the time interval of diastolic relaxation T_d^r is assumed to be one half of the systolic time interval T_s:

$$E(t) = \begin{cases} E_d + \frac{E_{es}-E_d}{2} \cdot \left\{1 - cos\left(\pi \frac{t}{T_s}\right)\right\} & 0 \le t \le T_s \\ E_d + \frac{E_{es}-E_d}{2} \cdot \left\{1 + cos\left(2\pi \frac{t}{T_s}\right)\right\} & T_s < t \le \frac{3}{2}T_s \\ E_d & \frac{3}{2}T_s < t \end{cases} \quad (9)$$

E_d and E_{es} are the diastolic elastance and end-systolic elastances. The parameters include the atrial and ventricular systole duration, T_S^a and T_S^v respectively. The P-R interval is defined here as the delay between the onset of the atrial and ventricular contraction T_{a-v}. The square root of the R-R interval length is T_{RR}.

$$T_S^a(s) = 0.2\sqrt{T_{RR}} \quad (10)$$

$$T_S^v(s) = 0.3\sqrt{T_{RR}} \quad (11)$$

$$T_{a-v}(s) = 0.12\sqrt{T_{RR}} \quad (12)$$

The cardiac pacemaker is represented using an Integral Pulse Frequency Modulation (IPFM) model, according to:

$$M(t) = \int_{t_{k-1}}^{t} m(t)dt = \int_{t_{k-1}}^{t} (m_0 + m_r(t))dt \quad (13)$$

In the equation above, $M(t)$ mimics the behavior of the transmembrane potential in the sino-atrial node It depends on the cumulative automaticity m_0, and the contribution of neural control input $m_r(t)$ since the end of the last heartbeat. A new heartbeat occurs at the time t_k when the transmembrane potential $M(t)$ reaches a predefined threshold potential $\Gamma = 1$.

$$\int_{t_{k-1}}^{t_k} m_r(t)dt = M(t_k) \ge \Gamma \quad (14)$$
$$t_k - t_{k-1} \ge 0.2(t_{k-1} - t_{k-2})$$

The function $m(t)$ is defined as the inverse of the instantaneous R-R interval $I(t)$:

$$m(t) = \frac{1}{I(t)} = \frac{1}{I_0 + \Delta I_{AB}(t)} \tag{15}$$

where I_0 is the nominal R-R interval, and $\Delta I_{AB}(t)$ is the control input from the arterial baroreflex control system.

B. Transcapillary Flow and Interstitial Fluid Volume

The transcapillary flow is solved analytically. The solution of the equations depends on the time constant ($\tau = RC = 4.6\ min$)) and the maximum interstitial volume change ($V_{max} = P_h C$). In the splanchnic venous region, the leg venous region and the abdominal venous region, the transcapillary flow is subtracted from the venous return. The following expressions define the fractions of interstitial volume and interstitial flow assigned to each compartment:

$$V^n(t) = \frac{P^n_{h,max}}{\sum_i P^i_{h,max}} \cdot V(t) \tag{16}$$

$$q^n(t) = \frac{P^n_{h,max}}{\sum_i P^i_{h,max}} \cdot q(t) \tag{17}$$

where $P^n_{h,max}$ is the maximum hydrostatic pressure in the nth compartment; and $\sum_i P^i_{h,max}$ is the sum of the maximum hydrostatic pressures of the three compartments.

C. Analytical Solutions for the Transcapillary Flow and the Interstitial Fluid Volume

Region I: Gradual increase in orthostatic stress over a period of length Δt.	$q(t) = \frac{V_{max}}{\Delta t} \cdot (1 - e^{-\frac{t}{\tau}})$ (18) $V(t) = V_{max} \cdot \left(\frac{t}{\Delta t} - \frac{\tau}{\Delta t}\left(1 - e^{-\frac{t}{\tau}}\right)\right)$ (19)
Region II: Full gravitational stress during period of duration T_{tilt}.	$q(t) = \frac{V_{max}}{\Delta t} \cdot (1 - e^{-\frac{\Delta t}{\tau}}) \cdot e^{-\frac{t-\Delta t}{\tau}}$ (20) $V(t) = V_{max} \cdot \left(1 - \frac{\tau}{\Delta t}\left(1 - e^{-\frac{\Delta t}{\tau}}\right) e^{-\frac{t-\Delta t}{\tau}}\right)$ (21)
Region III: Gradual decline in gravitational stress over a period of length Δt.	$q(t) = \frac{V_{max}}{\Delta t} \cdot \left(1 + \left(1 - e^{-\frac{\Delta t}{\tau}}\right) e^{-\frac{T_{tilt}}{\tau}}\right) \cdot e^{-\frac{t-(\Delta t + T_{tilt})}{\tau}} - \frac{V_{max}}{\Delta t}$ (22) $V(t) = V_{max} \cdot \left(1 - \frac{t - (\Delta t + T_{tilt})}{\Delta t}\right) - V_{max} \cdot \frac{\tau}{\Delta t} \cdot \left(1 - e^{-\frac{\Delta t}{\tau}}\right) \cdot e^{-\frac{T_{tilt}}{\tau}} +$ $+ V_{max} \cdot \frac{\tau}{\Delta t} \cdot \left(1 + \left(1 - e^{-\frac{\Delta t}{\tau}}\right) e^{-\frac{T_{tilt}}{\tau}}\right) \cdot \left(1 - e^{-\frac{t-(\Delta t + T_{tilt})}{\Delta t}}\right)$ (23)
Region IV: Post-orthostatic stress recovery of unspecified length.	$q(t) = -\frac{V_{max}}{\Delta t} \cdot \left(1 - e^{-\frac{\Delta t}{\tau}}\right) \cdot \left(1 - e^{-\frac{T_{tilt} + \Delta t}{\tau}}\right) \cdot e^{-\frac{t-(2\Delta t + T_{tilt})}{\tau}}$ (24) $V(t) = V_{max} \cdot \frac{\tau}{\Delta t} \cdot \left(1 - e^{-\frac{\Delta t}{\tau}}\right) \cdot \left(1 - e^{-\frac{T_{tilt} + \Delta t}{\tau}}\right) \cdot e^{-\frac{t-(2\Delta t + T_{tilt})}{\tau}}$ (25)

D. Initial Conditions

The following non-linear algebraic expressions are used to define the initial conditions. *C* indicates capacitance, *P* indicates pressure, and *R* indicates resistance. The subscripts indicate the compartments.

$$C_{ld}(P_{lvd} - P_{th}) - C_{ls}(P_{lvs} - P_{th}) = C_{rd}(P_{rvd} - P_{th}) - C_{rs}(P_{rvs} - P_{th}) \tag{26}$$

$$C_{ld}(P_{lvd} - P_{th}) - C_{ls}(P_{lvs} - P_{th}) = T^v_s \cdot \frac{P_{lvd}}{R_1} \tag{27}$$

$$T^v_s \cdot \frac{P_{lvd}}{R_1} = I_0 \cdot \left(\frac{P_1 - P_2}{R_2} + \frac{P_1 - P_6}{R_6}\right) \tag{28}$$

$$I_0 \cdot \frac{P_1 - P_2}{R_2} = I_0 \cdot \frac{P_2 - P_3}{R_3} \tag{29}$$

$$I_0 \cdot \frac{P_2 - P_3}{R_3} = I_0 \cdot \frac{P_3 - P_4}{R_{ub}} \tag{30}$$

$$I_0 \cdot \frac{P_3 - P_4}{R_{ub}} = I_0 \cdot \frac{P_4 - P_5}{R_4} \tag{31}$$

$$I_0 \cdot \frac{P_4 - P_5}{R_4} = I_0 \cdot \frac{P_5 - P_{ra}}{R_5} \tag{32}$$

$$I_0 \cdot \frac{P_1 - P_6}{R_6} = I_0 \cdot \frac{P_6 - P_7}{R_7} \tag{33}$$

$$I_0 \cdot \frac{P_6 - P_7}{R_7} = I_0 \cdot \left(\frac{P_7 - P_8}{R_8} + \frac{P_7 - P_{10}}{R_{10}} + \frac{P_7 - P_{12}}{R_{12}}\right) \tag{34}$$

$$I_0 \cdot \frac{P_7 - P_8}{R_8} = I_0 \cdot \frac{P_8 - P_9}{R_{rc}} \tag{35}$$

$$I_0 \cdot \frac{P_8 - P_9}{R_{rc}} = I_0 \cdot \frac{P_9 - P_{14}}{R_9} \tag{36}$$

$$I_0 \cdot \frac{P_{10} - P_{11}}{R_{sc}} = I_0 \cdot \frac{P_{11} - P_{14}}{R_{11}} \tag{39}$$

$$I_0 \cdot \frac{P_7 - P_{12}}{R_{12}} = I_0 \cdot \frac{P_{12} - P_{13}}{R_{lc}} \tag{40}$$

$$I_0 \cdot \frac{P_{12} - P_{13}}{R_{lc}} = I_0 \cdot \frac{P_{13} - P_{14}}{R_{13}} \tag{41}$$

$$I_0 \cdot \frac{P_{14} - P_{15}}{R_{14}} = I_0 \cdot \left(\frac{P_9 - P_{14}}{R_9} + \frac{P_{11} - P_{14}}{R_{11}} + \frac{P_{13} - P_{14}}{R_{13}}\right) \tag{42}$$

$$I_0 \cdot \frac{P_{14} - P_{15}}{R_{14}} = I_0 \cdot \frac{P_{15} - P_{ra}}{R_{15}} \tag{43}$$

$$I_0 \cdot \left(\frac{P_5 - P_{ra}}{R_5} + \frac{P_{15} - P_{ra}}{R_{15}}\right) = T^v_d \cdot \frac{P_{ra} - P_{rvd}}{R_{tri}} \tag{44}$$

$$T^v_d \cdot \frac{P_{ra} - P_{rvd}}{R_{tri}} = T^v_s \cdot \frac{P_{rvd} - P_{pa}}{R_{ro}} \tag{45}$$

$$T^v_s \cdot \frac{P_{rvd} - P_{pa}}{R_{ro}} = I_0 \cdot \frac{P_{pa} - P_{pv}}{R_{pv}} \tag{46}$$

$$I_0 \cdot \frac{P_{pa} - P_{pv}}{R_{pv}} = I_0 \cdot \frac{P_{pv} - P_{la}}{R_{li}} \tag{47}$$

$$I_0 \cdot \frac{P_{pv} - P_{la}}{R_{li}} = T^v_d \cdot \frac{P_{la} - P_{lvd}}{R_{mit}} \tag{48}$$

(49)

$$I_0 \cdot \frac{P_7 - P_{10}}{R_{10}} = I_0 \cdot \frac{P_{10} - P_{11}}{R_{sc}} \quad (37)$$

$$I_0 \cdot \frac{P_7 - P_{10}}{R_{10}} = I_0 \cdot \frac{P_{10} - P_{11}}{R_{sc}} \quad (38)$$

$$V_{total} - V_{total}^0 = \sum_{j \in \left\{ \substack{1,2,\ldots,10, \\ 12,15,ra,rv, \\ pa,pv,la,lv} \right\}} C_j \cdot \Delta P_j + \sum_{k \in \{11,13,14\}} \left[\frac{2V_{maxk}}{\pi} \cdot arctan\left(\frac{\pi C_{0k}}{2V_{maxk}} \cdot \Delta P_k \right) \right]$$

Acknowledgments

The authors would like to thank Richard Whittle for his assistance in model development. This research was funded by Lockheed Martin Corporation through the Independent Research and Development (IRAD) mechanism.

References

[1] "Acceleration in Aviation: G-Force," *Aerospace Medical Education Division, AAM-400*. 2021.

[2] K. Marshall-Goebel *et al.*, "Assessment of Jugular Venous Blood Flow Stasis and Thrombosis during Spaceflight," *JAMA Netw. Open*, vol. 2, no. 11, Nov. 2019, doi: 10.1001/jamanetworkopen.2019.15011.

[3] N. A. Vernice, C. Meydan, E. Afshinnekoo, and C. E. Mason, "Long-term spaceflight and the cardiovascular system," *Precis. Clin. Med.*, vol. 3, no. 4, pp. 284–291, Dec. 2020, doi: 10.1093/pcmedi/pbaa022.

[4] R. S. Whittle *et al.*, "Gravitational Dose-Response Curves for Acute Cardiovascular Hemodynamics and Autonomic Responses in a Tilt Paradigm," *J. Am. Heart Assoc.*, vol. 11, no. 14, pp. 1–18, Jul. 2022, doi: 10.1161/JAHA.121.024175.

[5] R. S. Whittle and A. Diaz-Artiles, "Gravitational effects on carotid and jugular characteristics in graded head-up and head-down tilt," *J. Appl. Physiol.*, vol. 134, no. 2, pp. 217–229, Feb. 2023, doi: 10.1152/japplphysiol.00248.2022.

[6] E. A. Hall, R. S. Whittle, and A. Diaz-Artiles, "Ocular perfusion pressure is not reduced in response to lower body negative pressure," *Npj Microgravity*, vol. 10, no. 1, p. 67, Jun. 2024, doi: 10.1038/s41526-024-00404-5.

[7] D. L. Harm *et al.*, "Invited Review: Gender issues related to spaceflight: a NASA perspective," *J Appl Physiol*, vol. 91, pp. 2374–2383, 2001.

[8] T. P. Stein and C. E. Wade, "The catecholamine response to spaceflight: role of diet and gender.," *Am. J. Physiol. Endocrinol. Metab.*, vol. 281, no. 3, pp. E500-6, Sep. 2001, doi: 10.1152/ajpendo.2001.281.3.E500.

[9] W. W. Waters, M. G. Ziegler, and J. V. Meck, "Postspaceflight orthostatic hypotension occurs mostly in women and is predicted by low vascular resistance.," *J. Appl. Physiol. Bethesda Md 1985*, vol. 92, no. 2, pp. 586–94, Feb. 2002, doi: 10.1152/japplphysiol.00544.2001.

[10] S. H. Platts *et al.*, "Effects of Sex and Gender on Adaptation to Space: Cardiovascular Alterations," *J. Womens Health*, vol. 23, no. 11, pp. 950–955, Nov. 2014, doi: 10.1089/jwh.2014.4912.

[11] J. M. Fritsch-Yelle, J. B. Charles, M. M. Jones, L. A. Beightol, and D. L. Eckberg, "Spaceflight alters autonomic regulation of arterial pressure in humans," *J. Appl. Physiol.*, vol. 77, no. 4, pp. 1776–1783, Oct. 1994, doi: 10.1152/jappl.1994.77.4.1776.

[12] J. M. Evans, C. F. Knapp, and N. Goswami, "Artificial gravity as a countermeasure to the cardiovascular deconditioning of spaceflight: Gender perspectives," *Front. Physiol.*, vol. 9, no. 7, Jul. 2018, doi: 10.3389/fphys.2018.00716.

[13] N. M. Arzeno, M. B. Stenger, S. M. C. Lee, R. Ploutz-Snyder, and S. H. Platts, "Sex differences in blood pressure control during 6° head-down tilt bed rest," *Am. J. Physiol.-Heart Circ. Physiol.*, vol. 304, no. 8, pp. H1114–H1123, Apr. 2013, doi: 10.1152/ajpheart.00391.2012.

[14] D. A. Ludwig, V. A. Convertino, D. J. Goldwater, and H. Sandler, "Logistic risk model for the unique effects of inherent aerobic capacity on +Gz tolerance before and after simulated weightlessness.," *Aviat. Space Environ. Med.*, vol. 58, no. 11, pp. 1057–61, Nov. 1987.

[15] J. M. Evans *et al.*, "Gender differences in autonomic cardiovascular regulation: spectral, hormonal, and hemodynamic indexes," *J. Appl. Physiol.*, vol. 91, no. 6, pp. 2611–2618, Dec. 2001, doi: 10.1152/jappl.2001.91.6.2611.

[16] S. Mark *et al.*, "The Impact of Sex and Gender on Adaptation to Space: Executive Summary," *J. Womens Health*, vol. 23, no. 11, pp. 941–947, Nov. 2014, doi: 10.1089/jwh.2014.4914.

[17] V. A. Convertino, "Gender differences in autonomic functions associated with blood pressure regulation," *Am. J. Physiol.-Regul. Integr. Comp. Physiol.*, vol. 275, no. 6, pp. R1909–R1920, Dec. 1998, doi: 10.1152/ajpregu.1998.275.6.R1909.

[18] P. Cowings, W. Toscano, D. Kanis, and F. Gebreyesus, "Individual Differences in the Temporal Profile of Cardiovascular Responses to Head Down Tilt and Orthostatic Stress With and Without Fluid Loading," presented at the NASA Human Research Program Investigator's Workshop, Feb. 12, 2013.

[19] J. R. Hordinsky, U. Gebhardt, H. M. Wegmann, and G. Schäfer, "Cardiovascular and biochemical response to simulated space flight entry.," *Aviat. Space Environ. Med.*, vol. 52, no. 1, pp. 16–8, Jan. 1981.

[20] J. M. Evans *et al.*, "Cardiovascular Regulation During Body Unweighting by Lower Body Positive Pressure," *Aviat Space Env. Med*, vol. 84, no. 11, pp. 1140–1146, Nov. 2013.

[21] R. S. Whittle, N. Keller, E. A. Hall, S. Patanam, B. J. Dunbar, and A. Diaz-Artiles, "Integrative cardiovascular dose–response to graded lower-body negative pressure," *Exp. Physiol.*, vol. n/a, no. n/a, doi: 10.1113/EP092483.

[22] M. Fois, A. Diaz-Artiles, S. Y. Zaman, L. Ridolfi, and S. Scarsoglio, "Linking cerebral hemodynamics and ocular microgravity-induced alterations through an in silico-in vivo head-down tilt framework," *Npj Microgravity*, vol. 10, no. 1, p. 22, Feb. 2024, doi: 10.1038/s41526-024-00366-8.
[23] A. D. Artiles, T. Heldt, and L. R. Young, "Effects of artificial gravity on the cardiovascular system: Computational approach," *Acta Astronaut.*, vol. 126, pp. 395–410, Sep. 2016, doi: 10.1016/j.actaastro.2016.05.005.
[24] R. S. Whittle and A. Diaz-Artiles, "Metabolic Modeling in Altered Gravity," in *IEEE Aerospace Conference*, 2020, pp. 1–17.
[25] R. Mirnezami, J. Nicholson, and A. Darzi, "Preparing for Precision Medicine," *N. Engl. J. Med.*, vol. 366, no. 6, pp. 489–491, Feb. 2012, doi: 10.1056/nejmp1114480.
[26] R. A. Gray and P. Pathmanathan, "Patient-specific cardiovascular computational modeling: Diversity of personalization and challenges," *J Cardiovasc. Transl. Res.*, vol. 11, no. 2, pp. 80–88, Apr. 2018, doi: 10.1007/s12265-018-9792-2.
[27] T. Heldt, "Computational Models of Cardiovascular Response to Orthostatic Stress," Massachusetts Institute of Technology, 2004.
[28] S. A. Zamanian, "Modeling and Simulating Human Cardiovascular Response to Acceleration," Massachusetts Institute of Technology, 2007.
[29] A. Diaz Artiles, "Exercise under Artificial Gravity-Experimental and Computational Approaches," Massachusetts Institute of Technology, 2015.
[30] R. S. Whittle, "Quantifying The Effects of Altered-Gravity and Spaceflight Countermeasures On Acute Cardiovascular And Ocular Hemodynamics," Texas A&M University, College Station, 2023.
[31] C. C. Gordon *et al.*, "2012 Anthropometric Survey of U.S. Army Personnel: Methods And Summary Statistics," U.S. Army Natick Soldier Research, Development and Engineering Center, Natick, Dec. 2014.
[32] M. Infeld *et al.*, "An approach towards individualized lower rate settings for pacemakers," *Heart Rhythm O2*, vol. 1, no. 5, pp. 390–393, Dec. 2020, doi: 10.1016/j.hroo.2020.09.004.
[33] S. Pfaffenberger *et al.*, "Size matters! Impact of age, sex, height, and weight on the normal heart size," *Circ. Cardiovasc. Imaging*, vol. 6, no. 6, pp. 1073–1079, Nov. 2013, doi: 10.1161/CIRCIMAGING.113.000690.
[34] H. Vellore, R. Galvan-Garza, L. Martin, and A. Diaz-Artiles, "Computational Modeling of Individual Differences in Cardiovascular Response during Parabolic Flight," in *IEEE Aerospace Conference*, IEEE Aerospace Conference, 2023.
[35] J. Feldschuh and Y. Enson, "Prediction of the Normal Blood Volume," *Circulation*, vol. 56, no. 4, pp. 605–611, Oct. 1977.
[36] P. Tikuisis, P. Meunier, and C. E. Jubenville, "Human body surface area: Measurement and prediction using three dimensional body scans," *Eur. J. Appl. Physiol.*, vol. 85, no. 3–4, pp. 264–271, 2001, doi: 10.1007/s004210100484.
[37] D. H. Horstman and S. M. Horvath, "Cardiovascular and temperature regulatory changes during progressive dehydration and euhydration," *J. Appl. Physiol.*, vol. 33, no. 4, pp. 446–450, Oct. 1972.
[38] J. Gonzalez-Alonso, R. Mora-Rodríguez, P. R. Below, and E. F. Coyle, "Dehydration markedly impairs cardiovascular function in hyperthermic endurance athletes during exercise," *J Appl Physiol*, vol. 82, no. 4, pp. 1229–1236, 1997.
[39] D. B. Dill and D. L. Costill, "Calculation of percentage changes in volumes of blood, plasma, and red cells in dehydration," *J. Appl. Physiol.*, vol. 37, no. 2, pp. 247–248, Aug. 1974.
[40] R. S. Whittle and A. Diaz-Artiles, "Modeling individual differences in cardiovascular response to gravitational stress using a sensitivity analysis," *J. Appl. Physiol.*, vol. 130, no. 6, pp. 1983–2001, Jun. 2021, doi: 10.1152/japplphysiol.00727.2020.
[41] M. A. Nystoriak and A. Bhatnagar, "Cardiovascular Effects and Benefits of Exercise.," *Front. Cardiovasc. Med.*, vol. 5, p. 135, 2018, doi: 10.3389/fcvm.2018.00135.
[42] A. Pelliccia, F. Culasso, F. M. Di Paolo, and B. J. Maron, "Physiologic Left Ventricular Cavity Dilatation in Elite Athletes," *Ann Intern Med*, vol. 130, no. 1, pp. 23–31, Jan. 1999.
[43] T. Bombardini, M. F. Costantino, R. Sicari, Q. Ciampi, L. Pratali, and E. Picano, "End-systolic elastance and ventricular-arterial coupling reserve predict cardiac events in patients with negative stress echocardiography," *BioMed Res. Int.*, vol. 2013, 2013, doi: 10.1155/2013/235194.
[44] M. R. Starling, R. A. Walsh, L. J. Dell'Italia, G. B. Mancini, J. C. Lasher, and J. L. Lancaster, "The relationship of various measures of end-systole to left ventricular maximum time-varying elastance in man.," *Circulation*, vol. 76, no. 1, pp. 32–43, Jul. 1987, doi: 10.1161/01.cir.76.1.32.
[45] R. D. Vanderbeek, "Period prevalence of acute neck injury in US Air Force pilots exposed to high G forces," *Aviat. Space Environ. Med.*, vol. 59, no. 12, pp. 1176–1180, 1988.
[46] C. A. Rickards and D. G. Newman, "G-induced visual and cognitive disturbances in a survey of 65 operational fighter pilots," *Aviat. Space Environ. Med.*, vol. 76, no. 5, pp. 496–500, May 2005.
[47] M. Shelhamer, "Parabolic Flight as a Spaceflight Analog," *J Appl Physiol*, vol. 120, pp. 1442–1448, 2016, doi: 10.1152/japplphysiol.01046.2015.-Ground.
[48] A. D. Artiles, D. Schor, and G. Clement, "Effects of inverted vision on hand-pointing performance in altered gravity during parabolic flight," in *2018 IEEE Aerospace Conference*, Big Sky, MT: IEEE, Mar. 2018, pp. 1–9. doi: 10.1109/AERO.2018.8396370.
[49] R. E. Klabunde, *Cardiovascular Physiology Concepts*. Wolters Kluwer Health/Lippincott Williams & Wilkins, 2012.
[50] S. K. Subramanian, V. K. Sharma, V. Arunachalam, R. Rajendran, and A. Gaur, "Comparison of Baroreflex Sensitivity and Cardiac Autonomic Function Between Adolescent Athlete and Non-athlete Boys – A Cross-Sectional Study," *Front. Physiol.*, vol. 10, p. 1043, Aug. 2019, doi: 10.3389/fphys.2019.01043.